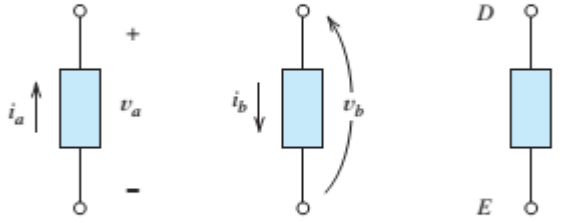
Tutorial Exercise

***P1.9.** The current through a given circuit element is given by $i(t) = 2e^{-t}$ A. As usual, time t is in seconds. Find the net charge that passes through the element in the interval from $t =$ 0 to $t = \infty$. (*Hint:* Current is the rate of flow of charge. Thus, to find charge, we must integrate current with respect to time.)

***P1.21.** Compute the power for each element shown in Figure P1.21. For each element, state whether energy is being absorbed by the element or supplied by it.



$-v_a = 30$ V, $i_a = -2$ A (a)

$v_b = -15$ V, $i_b = -4$ A (b)

$v_{ED} = 35$ V, $i_{DE} = 6$ A (c)

Figure P1.21

P1.24. Consider the element shown in Figure P1.24 which has $v(t) = -15$ V and $i(t) = 3e^{-2t}$ A. Compute the power for the circuit element. Find the energy transferred between $t = 0$ and $t = \infty$. Is this energy absorbed by the element or supplied by it?

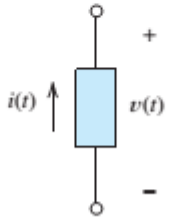


Figure P1.24